

Green Screens, Green Pixels and Green Shooting

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ABSTRACT

Sustainability and green producing are in high demand in all sectors of creative industries. Fortunately, this topic is received very well among film students providing an excellent opportunity for upcoming talent willing to apply new methods in creative processes. Virtualisation and Virtual Production in particular are predestined to play an essential role in fulfilling this demand. Factors that can be considered here are travel needs, lighting energy consumption, post-production complexity, energy sources and many more. The pandemic did propel these Virtual Production technologies to common practice, in particular large LED walls for In-Camera VFX (ICVFX). Some reports on the environmental impact of traditional film productions are available [albert 2020] estimating an average CO₂ demand of 2840 tonnes for tentpole film productions. However, these tentpole productions did not consider VFX. To date, there is little to no knowledge on the sustainability of Virtual Production and how it compares to traditional offline VFX productions. We take a closer look at two comparable productions, one using traditional offline rendering and post-production, the other using an LED wall and ICVFX. Energy requirements, creative opportunities and scalability are subjects of investigation and further discussion.

This abstract is a summary of a self published report on Virtual Production and its opportunities for sustainable film productions ¹.

KEYWORDS

Virtual Production, ICVFX, Green Shooting, Sustainability

ACM Reference Format:

Volker Helzle, Simon Spielmann, and Jonas Trotto. 2022. Green Screens, Green Pixels and Green Shooting. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH '22 Talks)*, August 07-11, 2022. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3532836.3536235>

1 INITIAL CONDITIONS

The productions compared here are the offline production “Sprout” from 2019 and the most recent production “Awakening” realized as a Virtual Production within the Set-Extension Workshop in 2021, an annual seminar at Filmakademie Baden-Württemberg. It involves students of diverse creative departments (Production, Set

¹<https://go.animationsinstitut.de/3g>

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SIGGRAPH '22 Talks, August 07-11, 2022, Vancouver, BC, Canada

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ACM ISBN 978-1-4503-9371-3/22/08.

<https://doi.org/10.1145/3532836.3536235>

Design, Directors of Photography, Lighting, Animation and VFX) learning to work within a green-screen set. Notice that the goal of the seminar is not necessary a fully produced film. Since 2020 this workshop has been realized using an LED wall. “Awakening” used a 4x10 meter curved LED wall featuring LED panels with a pixel pitch of 1.9mm². Pre- and post-production workstations as well as the displays power estimates were reduced by 30% to account for variations as students were also attending meetings and lectures.



Figure 1: On set the “Awakening” production

2 OFFLINE PRODUCTION “SPROUT”

The vast majority of power in this production was consumed in offline rendering for post-production. The studio recordings were realized in 2 days. The production had 8 shots with a total of 3233 frames including VFX. The estimate includes a usual amount of re-renders of the same shot. Our internal render management system³ keeps track of all jobs in a database. Jobs were executed on blades in our data center and on idle workstations in student and class rooms. The blades provide power consumption data via an internal meter. Workstations were measured using an off the shelf power meter⁴ and Cinebench R20 multi CPU benchmark⁵. We compared the measured data with spec sheets and system tools and found only minor deviations. Blades were calculated with 500 W each. The average render times were between 40 minutes and up to 2 hours. Workstations were estimated at 380 W. Pre-production (Previs, Techvis, Set Design) required 100 person days (8h a day). Post-production was accomplished within 300 person days. Displays were estimated with 80 W. Pre- and post-production involved

²https://www.leditgo.de/files/pdf/LEDitgo_rXone_Datenblatt.pdf

³<https://www.royalrender.de/>

⁴Dewenwils DHPM101A Energy Power Meter

⁵<https://www.maxon.net/en/cinebench>

5 students. As such the power consumption resulted in a total of 5073 kWh consisting of 4% pre-production, 13% post-production, 79% offline rendering and 4% for the displays. (figure 2 left).

3 LED WALL PRODUCTION "AWAKENING"

This production did not use a green-screen but a curved LED Wall to extend the real set by a virtual background. 8 shots were produced with a total of 8898 frames. Energy consumption was measured during 2 days and 17 hours on 2 high voltage power lines. This period included the entire production time consisting of 1 day setup and 2 days of production. Professional energy meters⁶ logged the current over time, from which we calculated the linear average power consumption at 4,6 kW for 65 hours. The total energy consumption for the LED wall was 299 kWh. One square meter of our 10x4 meter setup required approximately 115 W. The workstation providing the visuals for the LED wall had 2 Nvidia RTX A5000 graphics cards each running a resolution of 2560x2084 at 50Hz. Maximum power consumption for this system was 550 W. This resulted in 11 kWh power consumption when considering running for 20 hours straight at maximum capacity over 1 day setup and 2 days production. We double this value as an additional operator workstation was required. Pre-production required a higher demand on asset preparation as they needed to be final on the days of production in the studio. The workstations used in pre-production had recent graphics cards and their maximum system load was determined between 500 and 700 W. Notice that during pre-production the system will not run on maximum capacity all the time. Nevertheless we used the average of 600 W for the 5 weeks of pre-production involving 7 students estimating a total of 588 kWh. Post-production involved 6 persons for another 5 weeks resulting in 504 kWh. Notice that the total amount of frames is almost 3 times higher as in the offline production. As such the total power consumption for the LED wall production was estimated at 1594 kWh consisting of 37% pre-production, 31% post-production, 10% displays, 19% LED wall, 1% LED wall rendering and 2% offline rendering (figure 2 right).

4 CONCLUSION

Several aspects appear interesting to us. Most prominently, figure 2 shows that a Virtual Production can consume about a third of the energy needed for an comparable offline rendered production. Therefore, Virtual Production can be considered a sustainable and also green shooting solution. Furthermore, travel costs can be reduced as real sets can be digitized into virtual environments for LED volumes. In the discussed example, the LED Wall production produced more material (frames) in equal time on set compared to the offline production.

Both productions involved studio lighting which is remarkable when considering energy budgets. We did not include this into the energy calculation. Cooling and data storage energy consumption were also no factors of consideration at this point in time.

Apart from sustainability aspects, Virtual Production also allows for a unique opportunity of democratisation in filmmaking beyond shooting with LED walls. Tools for digital collaborative previsualisation, set design, lighting and shot planing are available at low costs[Spießmann et al. 2018] on consumer hardware.

Thereby, creativity and efficiency in general can be considered to be increased as post-production is no longer separated from the actual shoot. However, this comes with an increased demand on technical understanding in all departments and a willingness to adapt to new procedures and methodology. As this conclusion might sound like a clear vote for Virtual Production, we think that it is not a solution for all aspects of a film production. Therefore, traditional green-screen will remain part of our curriculum.

Lastly, we would like to consider scalability of the results as the LED wall was relatively small compared to professional studio spaces. Given the smaller scale of the offline student production ("Sprout") in terms of complexity and render times it also does not compare to a recent block buster production.

Hopefully, this report did send a clear signal towards the opportunities of new production technology and methods. However, questions arise as to common practice in major studios where average render times of up to 350 hours per frame are a reality[May 2021]. While this example is certainly not practice for all shots of a movie some questions remain: Is physical correctness really a requirement for an animated movie? Could something less energy hungry be equally visually impressive (e.g. by utilizing clever optimisation as common practice in game development due to hardware limitations)? Can smart uses of compositing achieve a somewhat similar result? The mindset to use all resources to their maximum capacity just because they are available should be reconsidered, facing the need to make film productions more sustainable.

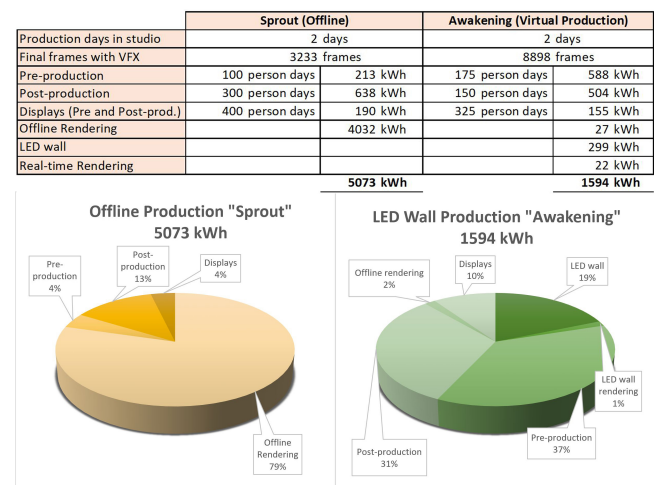


Figure 2: Offline and LED wall total power consumption

REFERENCES

- albert. 2020. *Screen New Deal*. Retrieved January 17, 2022 from <https://bit.ly/3GkyKoe>
- Steve May. 2021. *Pixar Special Event: How Pixar Sees Real Time Technologies Impacting Feature Animation Pipelines*. Retrieved January 17, 2022 from <https://realtimeconference.com/>
- Simon Spißmann, Volker Helzle, Andreas Schuster, Jonas Trottnow, Kai Götz, and Patricia Rohr. 2018. VPET: Virtual Production Editing Tools. In *ACM SIGGRAPH 2018 Emerging Technologies* (Vancouver, British Columbia, Canada) (SIGGRAPH '18). Association for Computing Machinery, New York, NY, USA, Article 22, 2 pages. <https://doi.org/10.1145/3214907.3233760>

⁶Fluke 1730, <https://www.fluke.com/>